

Problem

Given a string S, find the longest border of the string.

A border is a non-empty substring that is both a prefix and a suffix of the string, but it must be shorter than the entire string. S = "ababa"

Prefixes: "a", "ab", "aba", "abab" Suffixes: "a", "ba", "aba", "baba"

The longest common prefix and suffix is "aba".

Therefore, the answer is:

aba

Submissions

Input Format

A single line containing the string S.

Constraints

$1 \leq |S| \leq 10^5$ S contains lowercase English letters.

Output Format

Print the longest border of S.

Sample Input 0

ababa

Discussions

Sample Output 0

[Change Theme](#)

Language

Java 15



```

1  import java.io.*;
2  import java.util.Scanner;
3  public class Solution {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          if(!sc.hasNext()) return;
7          String s = sc.next();
8          char[] str = s.toCharArray();
9          int n = str.length;
10         int[] arr1 = new int[n];
11         int j=0;
12         for(int i=1;i<n;i++) {
13             while(j>0 && str[i]!=str[j]) {
14                 j=arr1[j-1];
15             }
16             if(str[i]==str[j]) {
17                 j++;
18             }
19             arr1[i]=j;
20         }
21         int borderLen = arr1[n-1];
22         if (borderLen>0) {
23             System.out.println(s.substring(0,borderLen));
24         }
25     }
26 }
```

Line: 26 Col: 2